Sarthak Agrawal

AI/ML Engineer (Agentic Systems)
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iamme1311.github.io/Portfolio

SUMMARY:

AI/ML engineer focused on designing agentic AI systems that autonomously diagnose, reason, and act in production environments. Experienced with tool-calling LLM agents, retrieval pipelines, structured reasoning loops, evaluation frameworks, and backend infrastructure using Python, FastAPI, LlamaIndex, and LangChain.

PROFESSIONAL SKILLS:

GenAI & Agentic Systems:
• LLM APIs (OpenAI, Gemini, Llama), prompt engineering, vector search, embeddings.
• Tool-calling agents, secure tool execution, structured reasoning, validation loops.
• RAG pipelines, knowledge curation, evaluator functions, guardrails & safety.

Cloud & DevOps:
•AWS (Lambda, S3, ECS, EKS, RDS, DynamoDB, CFT, CloudWatch), Docker, CI/CD, GitHub Actions.

ML Engineering:
• Model evaluation, monitoring, dataset curation, benchmark design.
• Latency optimization, caching, batching, model routing.

Backend Engineering:
• Python, FastAPI, Async programming, PostgreSQL, SQLAlchemy.
• System design, performance optimization, API security & RBAC.

WORK EXPERIENCE:

AI Engineer
RippleLinks, Bengaluru, India · June 2024 - Present

- Designed and deployed **agentic AI systems** enabling autonomous task execution (tool calling, routing, and multi-step reasoning) to replace manual workflows across operations.
- Designed backend services using **FastAPI + PostgreSQL + async workers**, enabling scalable orchestration of agents, tool APIs, and long-running tasks.
- Built **secure REST APIs** with OAuth2/JWT authentication, RBAC authorization, and request guardrails to ensure safe AI and data access.
- Implemented **agent action pipelines** with queue-based background execution, context caching, and self-correction loops, improving task throughput and reliability.
- Operationalized multiple AI providers (OpenAI / Ollama) in a **cost-optimized inference layer** with fallback, routing, and model selection policies.
- Delivered full **observability** via structured logging, traces, and performance metrics (OpenTelemetry + Grafana) resulting in faster debugging and reduced downtime.
- Containerized microservices using **Docker** and enabled **zero-downtime CI/CD deployments** with GitHub Actions, improving release frequency and stability.
- Optimized infra footprint and performance via async I/O, connection pooling, and request batching, reducing latency and compute costs.

Data Science Intern

Innomatics Research Labs, Hyderabad, India - January 2024 - April 2024

- Developed RAG pipeline using LangChain + Gemini with retrieval, prompt engineering, and validation loops.
- Built a semantic subtitle search engine using NLP + embeddings.
- Conducted ML experiments on 8.5k+ reviews dataset improving classification accuracy.

PROJECTS:

Multi-Agent Research Platform

Python · LangGraph · FastAPI · Anthropic Claude API · WebSockets · PostgreSQL

github.com/iamme1311/multi-agent-research

- Designed and deployed an autonomous multi-agent system using **LangGraph** where 4 specialized agents (Researcher, Analyst, Fact-Checker, Writer) collaborate to answer complex research queries via conditional routing and shared state.
- Built a real-time **Websocket endpoint** streaming per-agent reasoning traces to a split-pane web UI, enabling full observability of the agent decision loop.
- Implemented Tavily-powered web search for the Researcher agent and PostgreSQL session persistence for multi-turn query history.

RAG Pipeline + Evaluation Dashboard

Python · LlamaIndex · pgvector · RAGAS · FastAPI · Chart.js

github.com/iamme1311/rag-eval-dashboard

- Built a production RAG pipeline using **Llamaindex** with SentenceWindow chunking, Cohere embeddings, and **pgvector** for semantic retrieval over uploaded documents (PDF, DOCX, TXT).
- Integrated **RAGAS evaluation framework** to automatically score faithfulness, answer relevancy, and context precision per response, logged to PostgreSQL.
- Developed a Chart.js dashboard tracking quality metric trends over time with per-document breakdown and query history.

LLM Gateway + Observability Dashboard

Python · FastAPI· LiteLLM · PostgreSQL · Redis · Chart.js

github.com/iamme1311/llm-gateway

- Built an OpenAI-compatible proxy gateway routing requests to OpenAI, Antropic, and Gemini via **LiteLLM** with per-key rate limiting via Redis.
- Implemented request logging middleware capturing provider, model, token usage, latency, and cost per request into PostgreSQL.
- Developed a real-time observability dashboard showing daily cost per provider, latency p50/p95/p99, token trends, and sortable model comparison table.

EDUCATION:

Masters in Computer Applications - AI & ML - January 2022 - January 2024

Greater Noida, India

LANGUAGES:

English, Hindi, Tamil